u1982583 CH923 Test 28-10-19

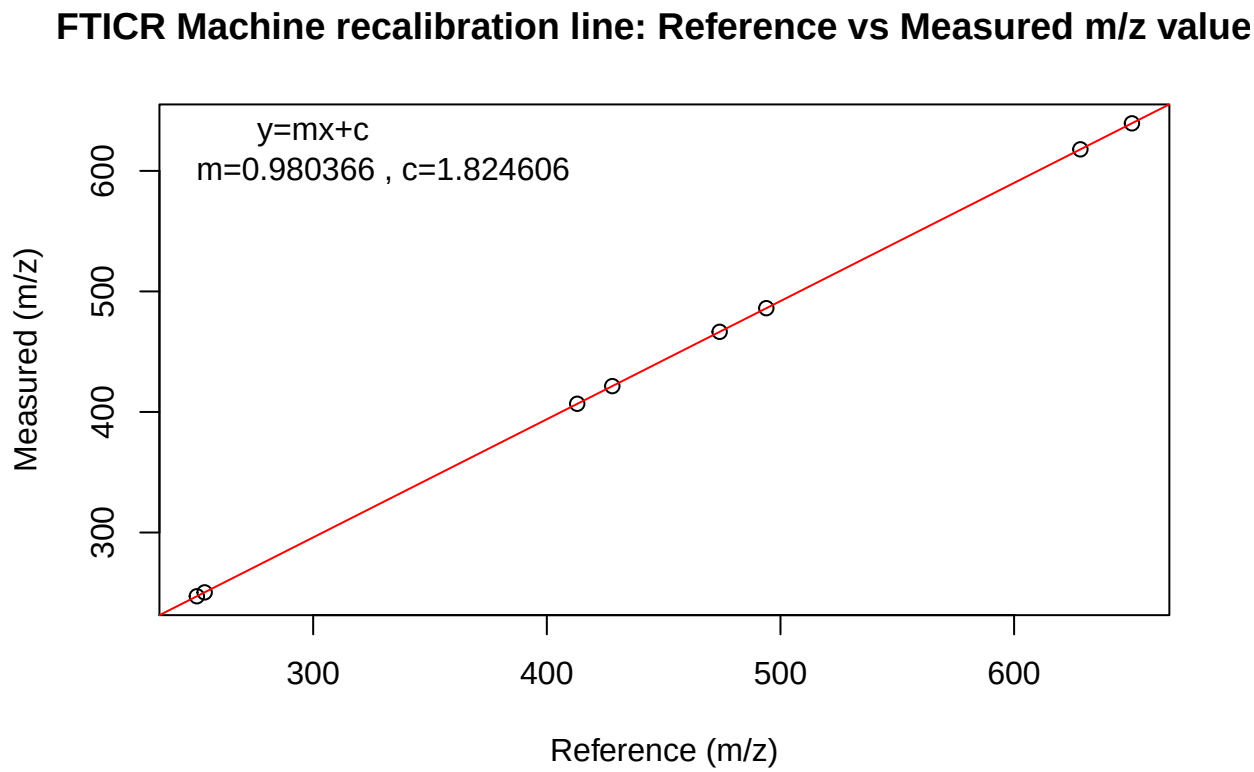
Question 2

Using only the values that have a response for a predictor, i.e trunckating the excel file we can fit a simple linear model as shown below. Then plot a scatterplot to show the relationship with the line of best fit shown according to the model. Fitting the linear model we get the following equation using the models coefficients : MeasuredPeak=0.980366 * Reference + 1.824606 Using the rearranged equation to get $Reference = \frac{(MeasuredPeak - 1.824606)}{0.980366}$ We can calculate the remaining values of x given the y as [1] 508.2753 [2] 965.9343 respectively for the last 2 values remaining.

```
library(readr)
dframe11<- read.csv("FTICR.csv")
attach(dframe11)
model11<- lm(MeasuredPeak~Reference)
summary(model11)
```

```
##
## Call:
## lm(formula = MeasuredPeak ~ Reference)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.066905 -0.021823  0.001936  0.025248  0.078860
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.8246060  0.0633024   28.82 1.16e-07 ***
## Reference   0.9803660  0.0001347 7278.94  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.053 on 6 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 5.298e+07 on 1 and 6 DF,  p-value: < 2.2e-16
```

```
Reference <-c(Reference)
Measured <-c(MeasuredPeak)
plot(Reference,Measured, xlab="Reference (m/z)", ylab = "Measured (m/z)", main="FTICR Machine recalibra
abline(model11, col="red")
text(330,600,"m=0.980366 , c=1.824606")
   text(300,630,"y=mx+c")
```




```
model11$coefficients
```

```
## (Intercept)   Reference
##    1.824606    0.980366
```

```
(500.1204-1.824606)/0.980366
```

```
## [1] 508.2753
```

```
(948.7938-1.824606)/0.980366
```

```
## [1] 965.9343
```

The precision of the estimated values is low as we have extrapolated the calibration curve to find out the new values. The estimation of the regression line is most accurate at $\bar{x}$, and becomes less accurate as you move away.Looking further at my equation: I can see that it varies by 1.824606 m/z values which shows a degree of variation in the readings compared with the reference which we know to be accurate and further shows that its precision is low so our calculations are less precice by that magnitude and potentially more as we dont know the relationship past our linear fit, could change to a curve relationship. Assumptions: 1.I assume that the fitted values and the residuals are independent, 2.errors have equal variances and and are normally distributed 3. errors are uncorrelated/independent.

Question 3

H_0: Left shoe size (cm) is the same as the the right shoe size (cm) H_1: The left shoe size (cm) is greater than the right shoe size (cm) Under H_0: T $\sim t_d$ so we reject H_0 at the 5% level if |T|>t_9(0.95)

with d being m-1=9

Do a F test for test of equal variances and see that the variances are equal, see the p value of 0.9639 > 0.05 so therefore accept H_0 that the variances are equal. Next do the paired t test that is paired according to the shoe pair and the alternative being left > right size.

```r
library(readr)
dframe22<- read_csv("Shoes.csv")
```

```
## Parsed with column specification:
## cols(
##   `Size of pair` = col_character(),
##   `Left (cm)` = col_double(),
##   `Right (cm)` = col_double()
## )
```

```r
attach(dframe22)
var.test(`Left (cm)`,`Right (cm)`)
```

```
##
##  F test to compare two variances
##
## data:  Left (cm) and Right (cm)
## F = 0.96941, num df = 9, denom df = 9, p-value = 0.9639
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.2407884 3.9028501
## sample estimates:
## ratio of variances
##          0.9694128
```

```r
t.test(`Left (cm)`, `Right (cm)`, var.equal = TRUE, alternative = "greater", paired = TRUE)
```

```
##
##  Paired t-test
##
## data:  Left (cm) and Right (cm)
## t = 2.0986, df = 9, p-value = 0.03263
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.02909959        Inf
## sample estimates:
## mean of the differences
##                    0.23
```

```r
qt(0.95, 9)
```

```
## [1] 1.833113
```

```r
1-pt(2.0986, 9)
```

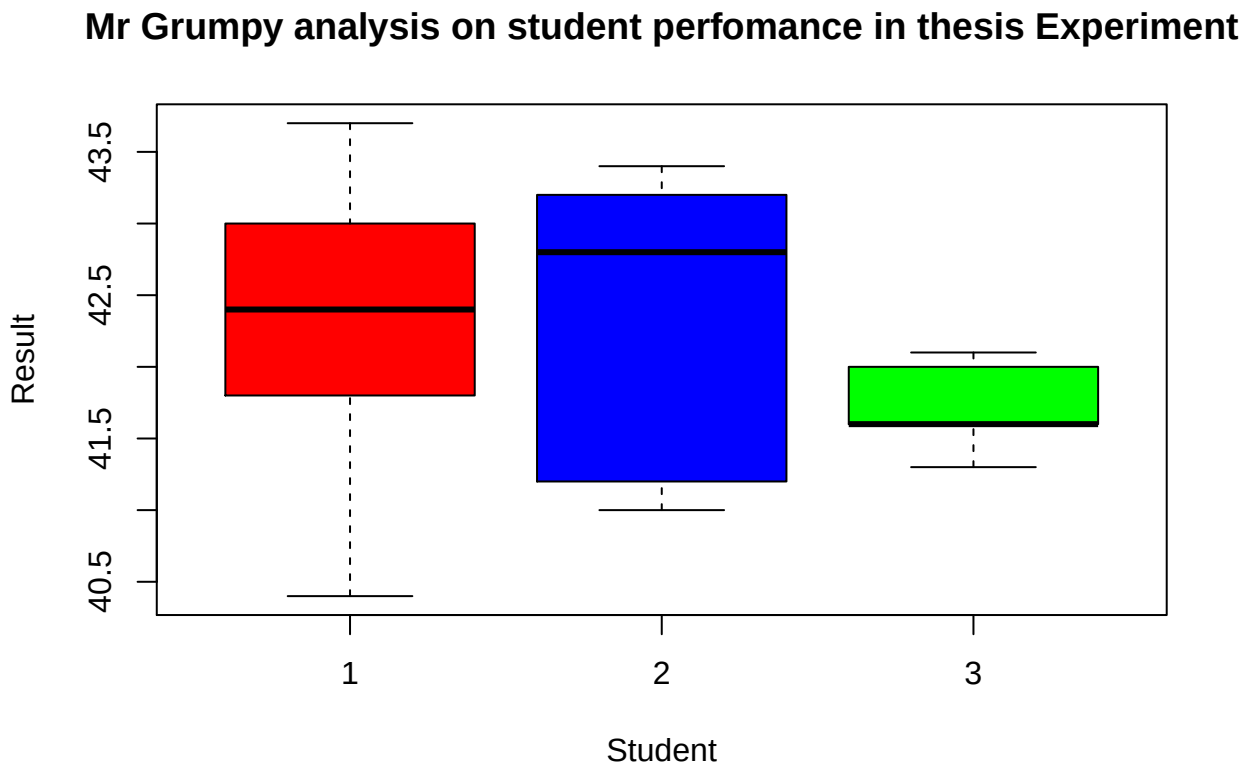
```
## [1] 0.03263319
```

Therefore T = 2.0986 and there was evidence to reject H0 at the 5% level as bigger than $t_9(0.95)$ of 1.833113. There was evidence that left pair is larger than the right pair. The p-value for this test was 0.03263 <0.05. The customers are therefore right and should demand a refund, however the number of samples (n) is low. Assumptions were: 1. Each population follows a normal distribution(Shapiro test, qqn plot for +/- skew needed) 2.The observations for each pair of shoes are independent to other pair of shoes 3. Samples are from populations with equal variances (Satisfied)

Question 4

H_0: All students are equal in their results H_1: At least one student is different in their result Under H_0: F ~ $F_{n-1,n-t}$ so we reject H_0 at the 5% level if F>$F_{n-1,n-t}(0.95)$ with the degrees of freedom being n-1=2 and n-t=12

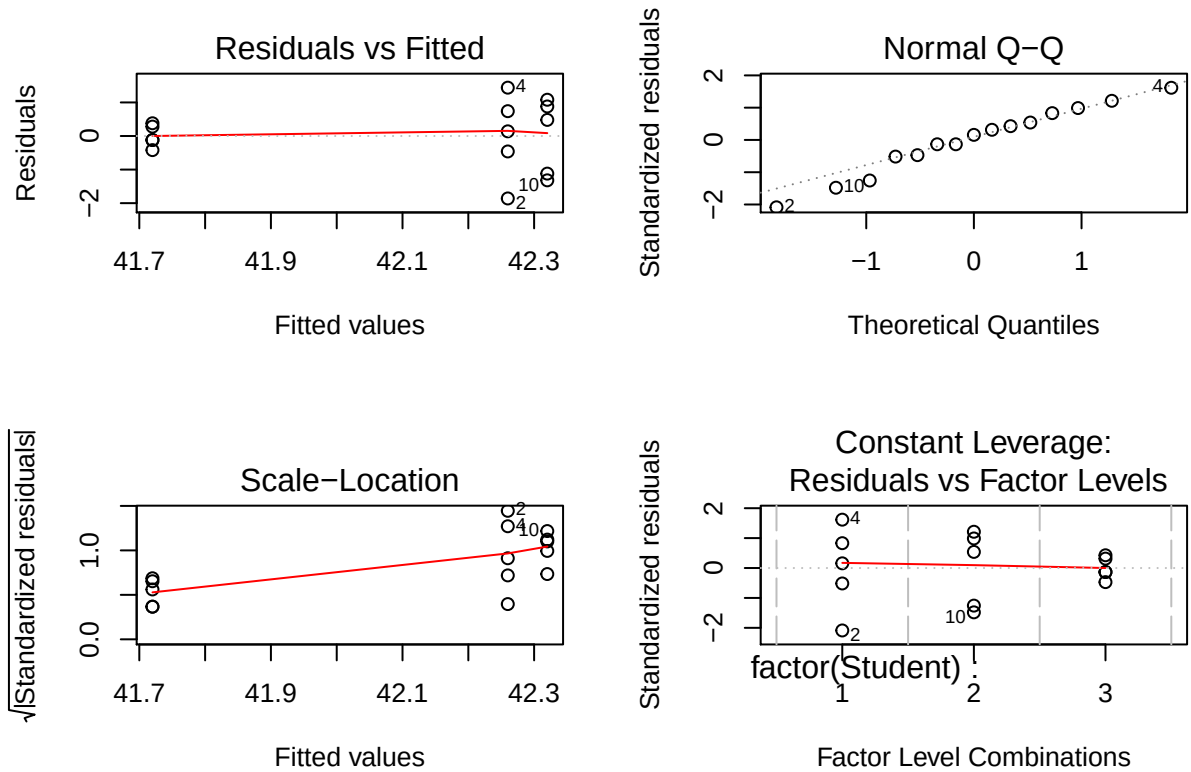
```
library(readr)
dframe33<-read.csv("Grumpy.csv")
attach(dframe33)
Students<-tapply(Result, Student, list)
boxplot(Students, xlab="Student", ylab="Result", main="Mr Grumpy analysis on student perfomance in thesi
```




```
anova1<-aov(Result ~ factor(Student))
summary(anova1)
```

```
##                 Df Sum Sq Mean Sq F value Pr(>F)
## factor(Student)  2  1.092  0.5460    0.55  0.591
## Residuals       12 11.908  0.9923
```

```
op<-par(mfrow=c(2,2))
plot(anova1)
```




```r
qf(0.95,2,12)
```

```
## [1] 3.885294
```

```r
1-pf(0.55, 2, 12)
```

```
## [1] 0.5908261
```

From the plot we can see equal variances with the widths of the plots being the same, showing student 3 being slightly lower than the other 2 students. Since this p-value is 0.591>0.05, there is no evidence to reject the H_0 that all of the treatments are the same. Our F value test statistic (F=0.55)is within the threshold of 3.885294 at the 5% confidence interval. Based on the data I believe this is true with litle differences in the means of the students (1 being 42.26, 2 being 42.32, 3 being 41.72) but the low number of repeats would prove to make this unreliable. Some of the anova assumptions are sattisfied: Relationship of residual to fitted values is shown to be linear with the plot(anova1) and the variances equal too with the points being spread out equally and qq plot showing normality, however a shapiro test for normaility is more of a quantitative test for normality.

Question 5

Chi-squared test for independence H0 : The attendance at the staff meeting and grade are independent. H1 : The attendance at the staff meeting and grade are are dependent. Having initially tested the chisquared expected values of the tabstats matrix, I realised row 2 and 3 need to be poole to satisfy the assumption of the test E_i<5. Having done that I can now do a chisquared test on the new pooled data tabstatpooled with 2 degrees of freedom. Under H0: $X = \Sigma_{i,j} \frac{(O_{i,j}-E_{i,j})^2}{E_{i,j}}$ $X \approx X_d^2$

where d = (r-1)(c-1) = (3-1)(2-1) = 2 Therefore we reject H0 at the 5% level if X > $X_2^2(0.95) = 5.99$ For these data X = 3.072 and so there is NO evidence to reject H0 at the 5% level. There is no evidence that attendance at the staff meeting is associated with grade. The p-value for this test was p = 0.2152

Assumptions:

1. Normally distributed
2. Ei >5 (Satisfied after pooling)

```r
tabstats<-matrix(c(14,17,4,17,4,5,0,11),4,2)
tabstats
```

```
##      [,1] [,2]
## [1,]   14    4
## [2,]   17    5
## [3,]    4    0
## [4,]   17   11
```

```r
chisq.test(tabstats)$expected
```

```
## Warning in chisq.test(tabstats): Chi-squared approximation may be incorrect
```

```
##           [,1]     [,2]
## [1,] 13.000000 5.000000
## [2,] 15.888889 6.111111
## [3,]  2.888889 1.111111
## [4,] 20.222222 7.777778
```

```r
tabstatspooled<-matrix(c(14,21,17,4,5,11),3,2)
tabstatspooled
```

```
##      [,1] [,2]
## [1,]   14    4
## [2,]   21    5
## [3,]   17   11
```

```r
chisq.test(tabstatspooled)$expected
```

```
##          [,1]     [,2]
## [1,] 13.00000 5.000000
## [2,] 18.77778 7.222222
## [3,] 20.22222 7.777778
```

```r
chisq.test(tabstatspooled)
```

```
##
##  Pearson's Chi-squared test
##
## data:  tabstatspooled
## X-squared = 3.072, df = 2, p-value = 0.2152
```

```r
qchisq(0.95,2)
```

```
## [1] 5.991465
```